

concept of bare metal installation. This means there is no need of any host operating system to install the hypervisor. By using this technology, there is no risk of corrupting the host operating system. These hypervisors are directly installed on the hardware without the need for any other operating system. On this hypervisor, multiple virtual machines are created.

A Type 1 hypervisor is a client hypervisor that interacts directly with hardware that is being virtualised. It is completely independent from the operating system, unlike a Type 2 hypervisor, and boots before the OS. Currently, Type 1 hypervisors are being used by all the major players in the desktop virtualisation space, including but not limited to VMware, Microsoft and Citrix.

Type 2 (hosted) hypervisors execute within a conventional operating system environment. With the hypervisor layer as a distinct second software level, guest operating systems run at the third level above the hardware. A Type 2 hypervisor is a type of client hypervisor that sits on top of an operating system. Unlike Type 1, a Type 2 hypervisor relies heavily on the operating system. It cannot boot until the operating system is already up and running and, if for any reason the operating system crashes, all end users are affected. This is a big drawback of Type 2 hypervisors, as they are only as secure as the operating system on which they rely. Also, since Type 2 hypervisors depend on an OS, they are not in full control of the end user's machine.

Hypervisors in the industry

The table below gives a few examples of hypervisors and the corresponding cloud service provider companies.

Hypervisor	Cloud service provider
Xen	Amazon EC2 IBM SoftLayer Fujitsu Global Cloud Platform Linode OrionVM
ESXi	VMware Cloud
KVM	Red Hat HP Dell Rackspace
Hyper-V	Microsoft Azure

Cloud service providers and what they offer

Currently, there are a number of cloud service providers in the global market. Here are a few of them in the storage domain:

- JustCloud
- Zipcloud
- Dropbox
- Zoolz
- Livedrive, etc.

Research areas in cloud computing

Cloud computing and related services are very frequently taken up for further research by scholars as well as academicians. As cloud services have a number of domains, deployment models and

Type 1 hypervisors

- VMware ESXi
- Citrix Xen
- KVM (Kernel Virtual Machine)
- Hyper-V

Type 2 hypervisors

- VMware Workstation
- VirtualBox

respective algorithmic approaches, there is a huge scope for research.

The following topics offer a lot of scope for research scholars in the cloud infrastructure domain:

- Energy optimisation
- Load balancing
- Security and integrity
- Privacy in multi-tenancy clouds
- Virtualisation
- Data recovery and backup
- Data segregation and recovery
- Scheduling for resource optimisation
- Secure cloud architecture
- Cloud cryptography
- Cloud access control and key management
- Integrity assurance for data outsourcing
- Verifiable computation
- Software and data segregation security
- Secure management of virtualised resources
- Trusted computing technology
- Joint security and privacy-aware protocol design
- Failure detection and prediction
- Secure data management within and across data centres
- Availability, recovery and auditing
- Secure computation outsourcing
- Secure mobile cloud

Problem formulation for R&D in cloud computing

Problem formulation is the first step that needs to be taken to work on any research task, whether it is related to a thesis or a corporate research project. Once the problem is found after investigating different research papers and media reports, the work can be started.

First, research scholars or academicians need to focus on their own interests and skills. If a researcher is very good in mathematical modelling, then the research project selected ought to be in cloud cryptography, power optimisation, energy optimisation, security or any topic that has mathematical functions. In case of a strong background in algorithm design, research work can be done in scheduling, trusted architectures, load balancing or related fields. It often happens that the research topic is selected without analysing one's own expertise and interests.

Once the research topic is selected, then there is the task of literature survey, without which no genuine research work can be done. In this process, research papers from prominent international journals are analysed in relation to the specific area that is selected for research. If the topic is energy optimisation, then the research papers should be related to this particular topic